

Quiz 3: MA 5755, Department of Mathematics, IIT Madras

Course name : *Data Analysis & Visualization in R/Python/SQL*

Max. Marks: 30

Roll No:

Name:

A

- Write your roll number and name. Nothing else is needed.

1. Which statement best describes a discriminative model?
 - (a) It models how the data is generated from latent variables
 - (b) It models the joint distribution of inputs and outputs
 - (c) **It directly models the relationship between inputs and class labels**
 - (d) It ignores the labels completely
2. In LDA, the assumption of a common covariance matrix implies that:
 - (a) All classes have identical means
 - (b) **Decision boundaries are linear functions of the input**
 - (c) Classes are independent
 - (d) The model is nonparametric
3. Logistic regression differs from LDA primarily because:
 - (a) It produces nonlinear boundaries
 - (b) **It does not assume a distribution for X**
 - (c) It cannot be regularized
 - (d) It ignores class labels
4. The log-odds representation in logistic regression implies that:
 - (a) Probabilities are linear in x
 - (b) **Log-odds are linear in x**

- (c) Outputs are binary
 - (d) Loss is quadratic
5. A practical consequence of logistic regression focusing on $P(Y|X)$ is:
- (a) It uses more information than LDA
 - (b) **It is more robust to violations of distributional assumptions**
 - (c) It is always more accurate
 - (d) It ignores regularization
6. Regularization in logistic regression is primarily used to:
- (a) Increase bias only
 - (b) Reduce sample size
 - (c) **Control model complexity and prevent overfitting**
 - (d) Eliminate features automatically in all cases
7. L1 regularization is particularly useful because:
- (a) It smooths coefficients
 - (b) **It can produce sparse models with feature selection**
 - (c) It increases variance
 - (d) It removes the loss function
8. The margin in SVM represents:
- (a) Training error
 - (b) Distance between class means
 - (c) **Distance between the boundary and closest data points**
 - (d) Kernel width

9. Maximizing the margin tends to:

- (a) Increase overfitting
- (b) **Improve robustness and generalization**
- (c) Reduce model complexity to zero
- (d) Eliminate noise

10. Slack variables in SVM are introduced to:

- (a) Increase dimensionality
- (b) Remove constraints
- (c) **Allow violations of the margin in nonseparable data**
- (d) Normalize features

11. A key computational property of SVM is that:

- (a) It depends on raw features only
- (b) **It depends only on inner products between data points**
- (c) It avoids optimization
- (d) It uses clustering

12. The kernel trick allows us to:

- (a) Work in high-data size
- (b) **Work in high-dimensional feature spaces without explicit mapping**
- (c) Eliminate regularization
- (d) Avoid labels

13. Using kernels results in:

- (a) Always linear models

- (b) No dependence on data
- (c) **Nonlinear decision boundaries in input space**
- (d) Deterministic outputs

14. Support Vector Regression differs from standard regression because:

- (a) It uses squared loss only
- (b) **It ignores small residuals within a tolerance region**
- (c) It removes regularization
- (d) It assumes Gaussian noise

15. A large ϵ in SVR typically leads to:

- (a) Overfitting
- (b) Complex models
- (c) **Simpler, smoother functions**
- (d) Higher variance

16. The unified learning framework emphasizes:

- (a) Data preprocessing only
- (b) **Loss function, regularization, and function space**
- (c) Kernel choice only
- (d) Optimization only

17. Changing the loss function primarily affects:

- (a) Input dimension
- (b) **How errors are penalized**
- (c) Feature scaling

(d) Kernel computation

18. LDA produces:

(a) Quadratic boundary

(b) Nonlinear boundary

(c) **Linear boundary**

(d) Tree structure

19. Slack variable ξ_i measures:

(a) **Violation of margin**

(b) Distance to origin

(c) Variance

(d) Kernel

20. Squared loss is used in:

(a) Classification only

(b) Classification and Regression both

(c) Neither classification nor regression

(d) **Regression only**

21. In LDA, replacing the common covariance Σ with class-specific covariances Σ_k leads to:

(a) Linear decision boundaries with higher variance

(b) Identical decision boundaries

(c) **Quadratic decision boundaries due to class-dependent scaling**

(d) No change in the classifier

22. Suppose all training points lie far from the decision boundary in an SVM. What is the implication?
- (a) All points become support vectors
 - (b) The model becomes unstable
 - (c) **Most points have zero influence on the solution**
 - (d) The margin collapses
23. Which scenario most strongly favors LDA over logistic regression?
- (a) **Approximately Gaussian class distributions**
 - (b) Highly nonlinear boundaries
 - (c) Presence of strong outliers
 - (d) High-dimensional sparse data
24. In the dual formulation of SVM, why is the solution expressed in terms of inner products?
- (a) To reduce dimensionality
 - (b) To eliminate constraints
 - (c) **Because the optimal solution lies in the span of training data**
 - (d) To avoid regularization
25. In kernel methods, why is the choice of kernel critical?
- (a) **It implicitly defines the geometry and hypothesis space of the model**
 - (b) It determines optimization speed only
 - (c) It affects only regularization
 - (d) It removes dependence on data